

sintaxismark: English Documentation

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1. Overview

sintaxismark is a LaTeX package for annotating syntactic constituents, grammatical functions, heads, and spans directly over running text. It is intended for teaching materials, grammar handouts, slides, compact linguistic examples, and analyses where a full syntactic tree would be visually too heavy.

The package has two complementary modes: **classic mode**, based on TikZ overlays and commands such as `\spanline`, `\openbox`, `\tagbelow`, `\SES`, `\PVS`, etc.; and **grid mode**, an experimental LuaLaTeX-only mode that reads nested markup, lays out tokens on a grid, and prevents diagonal lines in multiline examples.

2. Requirements

Classic mode requires TikZ, xparse, xcolor, expl3, and varwidth. Grid mode additionally requires LuaLaTeX, because layout computation is performed in Lua. With gb4e, load gb4e before sintaxismark.

3. Loading the package

```
\usepackage{sintaxismark}
\usepackage[docencia]{sintaxismark}
\usepackage[impresion]{sintaxismark}
\usepackage[diapositivas]{sintaxismark}
\usepackage[publicacion]{sintaxismark}

\setsintaxisschema{diapositivas}
```

4. Classic mode

Core commands

```
\openbox{OD}{Juan bought apples}
\spanline{SES}{Juan} \spanline{PVS}{bought apples}
\tagbelow{NN}{Juan}
\tagabove{ST}{Juan bought apples}
\synul{bought}
```

Frequent shortcuts

Command	Label
<code>\SES{...}</code>	simple subject
<code>\ST{...}</code>	tacit subject
<code>\SEC{...}</code>	compound subject
<code>\PVS{...}</code>	simple verbal predicate
<code>\PVC{...}</code>	compound verbal predicate
<code>\OS{...}</code>	simple sentence
<code>\OD{...} / \OD*{...}</code>	direct object: tag/underline or box
<code>\OI{...} / \OI*{...}</code>	indirect object: tag/underline or box
<code>\MI{...} / \MI*{...}</code>	indirect modifier: tag or box
<code>\MD{...}</code>	direct modifier
<code>\N{...}, \NN{...}, \NV{...}</code>	heads / nuclei

Classic graphical keys

Key	Function
color	annotation color

linew	line width
liney	vertical position of upper span lines
linegap	gap between text and upper vertical ends
labsep	separation between upper line and label
boxd	depth of lower open brackets
boxsep	separation between lower bracket and label
tagy / tagx	vertical/horizontal position of floating tags
leftend / rightend	show or hide vertical ends
leftarrow / rightrightarrow	add arrowheads

5. Long examples in classic mode

For long examples, use `SMlinewrap`, no-break variants, or manual splitting.

```
\SMlinewrap{The \MI{department of some deteriorated blocks} was on the ground floor.}
\openboxNB{MI}{department of some deteriorated blocks}
\spanlineNB{SES}{The department of some deteriorated blocks}
\breakspan
  {\SES{\MI{The department of some blocks}}}
  {\SES{\MI{was on the ground floor}}}
```

6. Experimental grid mode

Grid mode does not draw each mark immediately. It first reads the entire `sintaxisgrid` environment, extracts tokens, infers ranges, computes line breaks, and draws the full analysis. Code indentation is only visual: commands and braces provide the structure.

Basic example

```
\begin{sintaxisgrid}[width=10cm]
\SA{
  \SES{Juan}
  \PVS{
    saw
    \OD*{
      \MD{a}
      \N{tiger}
    }
  }
}
\end{sintaxisgrid}

% compact equivalent:
\begin{sintaxisgrid}[width=10cm]
\SA{\SES{Juan}\PVS{saw \OD*{\MD{a}\N{tiger}}}}
\end{sintaxisgrid}
```

Command interpretation in grid mode

`SES`, `ST`, `SEC`, `PVS`, `PVC`, and `OS` produce upper spans. Starred forms such as `OD*`, `MI*`, and `COMP*` produce lower boxes. Unstarred `OD` and `OI` produce lower underlined functions. `MD`, `N`, `NN`, `NV`, and similar labels produce simple lower tags.

Use with gb4e

```
\begin{exe}
\ex
\begin{sintaxisgrid}[width=\linewidth]
\SA{\SES{Juan}\PVS{saw \OD*{\MD{a}\N{tiger}}}}
\end{sintaxisgrid}
\end{exe}
```

Global grid options

```
\sintaxismarksetup{
  grid/lower-level-gap=0.75,
  grid/lower-label-sep=0.18,
  grid/token-pad-left=0.06,
```

```

    grid/token-pad-right=0.06
}

```

Global key	Function
grid/width	default width
grid/valign	vertical alignment: base, t, c, b
grid/yoffset	manual vertical correction
grid/upper-y-base	distance between text and upper vertical ends
grid/upper-leg	length of upper vertical ends
grid/upper-level-gap	separation between upper levels
grid/upper-label-sep	separation between upper line and label
grid/lower-y-base	distance to lower zone
grid/lower-level-gap	separation between lower levels
grid/lower-leg-top	starting height of lower vertical ends
grid/lower-label-sep	separation between lower box and label
grid/tag-y-base	base distance of simple lower tags
grid/tag-level-gap	separation between tag levels
grid/token-gap	horizontal separation between tokens
grid/token-pad-left/right	space before/after first/last token
grid/linegap, grid/line-base, grid/line-lower- vertical	vertical line spacing

Local environment options

```

\begin{sintaxisgrid}[width=10cm, lower-level-gap=0.50, token-pad-left=0.03]
...
\end{sintaxisgrid}

```

Local mark options

```

\begin{sintaxisgrid}[width=10cm]
\SA{
  \SES[color=blue]{Juan}
  \PVS[color=black, arrow=right]{
    saw
    \OD*[color=red, no-leftend]{
      \MD[color=gray]{a}
      \N{tiger}
    }
  }
}
\end{sintaxisgrid}

```

Local option	Function
color=<color>	line and label color
linecolor / line-color	line color
labelcolor / label-color	label color
textcolor / text-color	alias for label color
leftend=false / rightend=false	remove left/right vertical end
no-leftend / no-rightend / no-ends	aliases for missing ends
arrow=left/right/both/none	arrowheads per mark
leftarrow / rightrightarrow / leftrightarrow / no-arrow	arrow aliases

7. Practical notes

Use LuaLaTeX with sintaxisgrid. Compile twice when using classic overlays. In grid mode, indentation is not structural; only commands and braces are structural.